

Comprehensive Model Comparison Report

Genetic Variant Pathogenicity Prediction

4 Model Types | 2 Training Modes | 2 Feature Engineering States
32 Total Configurations | Optuna Bayesian Optimization

Models: LightGBM, XGBoost, Neural Network (MLP), Deep Neural Network
Dataset: 4,287 Genetic Variants | 119 Features | 4 Gene Panels

Best Result: LightGBM - F1 = 0.7901

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1. Methodology

1.1 Experimental Design

A full factorial experiment was conducted to compare 4 model architectures, 2 feature engineering approaches, and 2 training modes across 4 gene panels, resulting in 32 unique configurations. Each configuration was independently optimized using Bayesian hyperparameter search (Optuna) with stratified cross-validation, and evaluated on a held-out 20% test set.

1.2 Models

LightGBM: Gradient boosting with leaf-wise tree growth and native categorical feature support. 200 Optuna trials, 5-fold StratifiedKFold CV. Hyperparameters: num_leaves (10-120), max_depth (3-15), learning_rate (0.005-0.3), n_estimators (50-800), min_child_samples (3-50), reg_alpha, reg_lambda, colsample_bytree (0.3-1.0), subsample (0.3-1.0), max_bin (63-511).

XGBoost: Gradient boosting with level-wise growth and L1/L2 regularization. 200 Optuna trials, 5-fold CV. Hyperparameters: max_depth (3-15), learning_rate (0.005-0.3), n_estimators (50-800), gamma (0-5), min_child_weight (1-10), subsample (0.3-1.0), colsample_bytree (0.3-1.0), scale_pos_weight for class imbalance.

Neural Network (NN): Shallow MLP with 1-2 hidden layers, Batch Normalization, ReLU activation, and Dropout. PyTorch + DirectML (AMD GPU). 100 Optuna trials, 3-fold CV. Hyperparameters: hidden_dim (32-256), dropout (0.1-0.5), lr (1e-4 to 1e-2), weight_decay (1e-5 to 1e-2), batch_size (32/64/128). BCEWithLogitsLoss with pos_weight, early stopping (patience=15, max 150 epochs).

Deep Neural Network (DNN): Deep MLP with 3-5 hidden layers and residual connections every 2 layers. Stronger regularization: dropout (0.2-0.6), ReduceLROnPlateau scheduler. 100 Optuna trials, 3-fold CV. Hyperparameters: hidden_dims (64-512), n_layers (3-5).

1.3 Feature Engineering

No-FE (Baseline): The original dataset has 119 columns. The following columns are dropped: 4 leakage columns (clinvar_sig, clinvar_id, Germline review status, Stars), 9 high-cardinality ID columns (base_cchange, base_achange, base_chrom, base_pos, base_ref_base, base_alt_base, ref_amino, alt_amino, alphamissense_protein_variant), and 1 target column. This leaves 99 original numeric/categorical features, including all prediction scores and rankscores from 20+ bioinformatics tools (SIFT, PolyPhen2, CADD, REVEL, MetaSVM, etc.).

Additionally, 6 sequence text columns are removed as raw text but re-encoded as numeric features via character-level 3-gram (trigram) frequency using CountVectorizer(analyzer="char", ngram_range=(3,3), max_features=200, min_df=3).

The per-column k-mer counts are:

```
Ref_Sequence:      64 trigrams (DNA, 4-letter alphabet -> max 64 possible)
Alt_Sequence:      64 trigrams
DNA_11mer_Ref:     64 trigrams
DNA_11mer_Alt:     64 trigrams
Prot_11mer_Ref:    200 trigrams (protein, 20-letter alphabet -> capped at 200)
Prot_11mer_Alt:    200 trigrams
TOTAL K-MER:       656 new features
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Final no-FE feature count: 99 original + 656 k-mer = ~755 features (exact count varies slightly by panel filtering and multi-panel one-hot encoding). This is significantly more than the ~99 base features because the k-mer encoding expands each sequence column into hundreds of numeric frequency features.

With-FE: Bioinformatics-driven feature extraction replacing raw scores and sequences: Grantham Distance and Category (amino acid physicochemical difference), BLOSUM62 substitution score, CpG site detection, GC content (11-mer window), transition vs transversion classification, gnomAD-derived features (log10_af, allele frequency bin, is_absent flag, homozygote ratio), conservation product (PhyloP x PhastCons), and tool consensus score (ratio of 10 damaging predictions). Correlation filter ($r > 0.99$) to remove redundant features. Total features: ~53-57.

1.4 Training Modes

Single-Panel: Separate model trained per panel (General, Hereditary_Cancer, PAH). CFTR excluded (100% pathogenic - no negative class). Panel column dropped.

Multi-Panel: Single unified model on all 4,287 variants. Panel membership one-hot encoded as features. CFTR samples included.

1.5 Evaluation

Train/test split: 80/20 stratified (random_state=42). 9 metrics computed: F1, AUC-ROC, AUC-PR, MCC, Precision, Recall, Specificity, Balanced Accuracy, Cohen's Kappa. Post-hoc threshold optimization: sweep 0.10-0.90 in steps of 0.01 to maximize F1 on test set. Both default (0.5) and optimized threshold F1 scores recorded.

2. Dataset Overview

Source: open_cravat_curation_v2_imputed.csv
Samples: 4,287 genetic variants
Original features: 119 columns
Target: Binary classification (Benign vs Pathogenic)
Class distribution: 68% Benign (2,920), 32% Pathogenic (1,367)

Gene Panel Distribution

General:	3,156 variants	(73.6%)
Hereditary_Cancer:	715 variants	(16.7%)
PAH:	324 variants	(7.6%)
CFTR:	92 variants	(2.1%) - 100% pathogenic

Feature categories include: pathogenicity prediction scores from 20+ bioinformatics tools (SIFT, PolyPhen2, CADD, REVEL, etc.), population allele frequencies (gnomAD), conservation scores (PhyloP, PhastCons), amino acid physicochemical properties (molecular weight, polarity, chirality), and DNA/protein sequence data (6 columns).

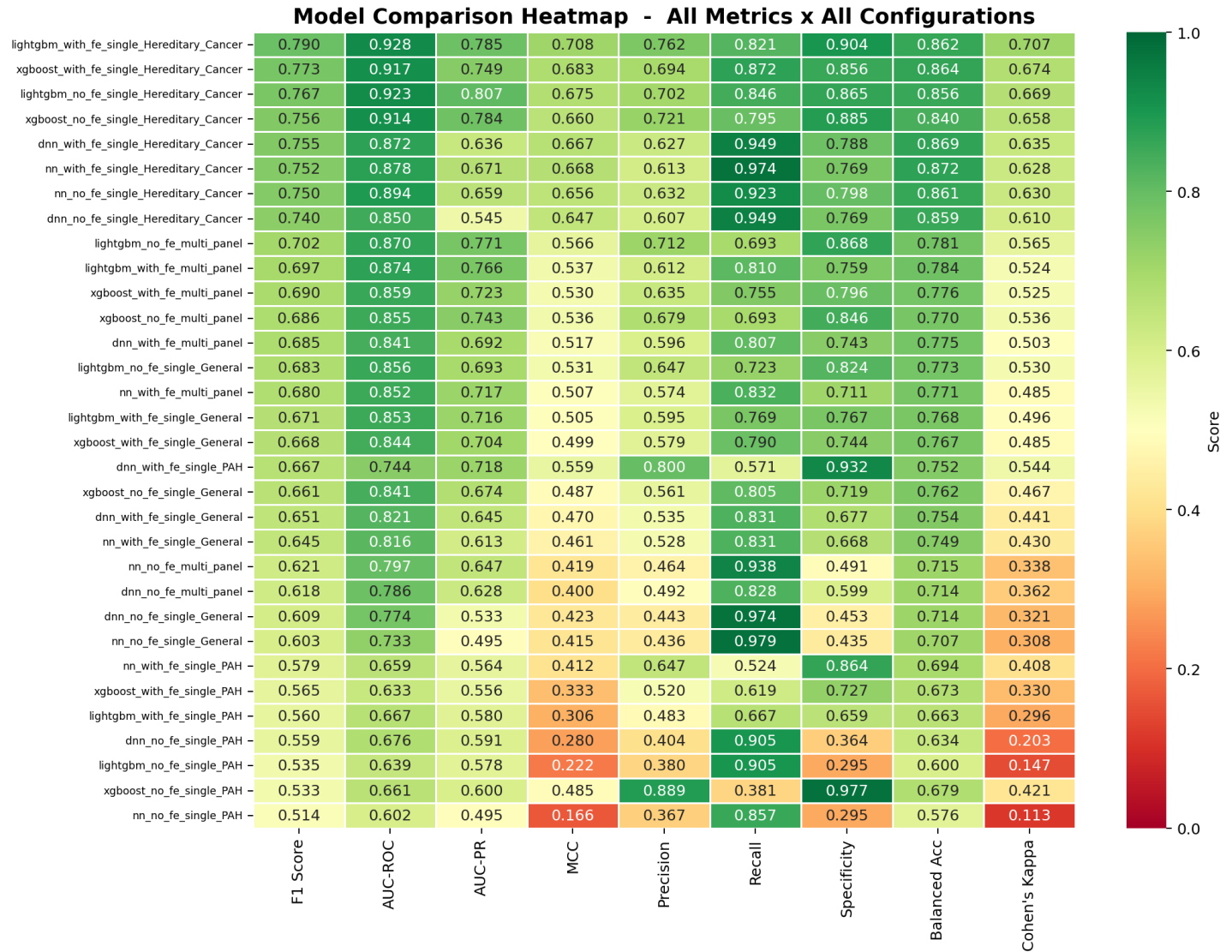
3. Results - All 32 Configurations

Ranked by F1 score. Top 5 highlighted green, bottom 5 highlighted red.

#	Model	Panel	Mode	FE	F1	AUC-ROC	MCC	Prec	Recall	Spec	Thresh
1	LightGBM	Hereditary_Cancer	Single	Yes	0.7901	0.9283	0.7083	0.7619	0.8205	0.9038	0.62
2	XGBoost	Hereditary_Cancer	Single	Yes	0.7727	0.9172	0.6827	0.6939	0.8718	0.8558	0.41
3	LightGBM	Hereditary_Cancer	Single	No	0.7674	0.9233	0.6746	0.7021	0.8462	0.8654	0.60
4	XGBoost	Hereditary_Cancer	Single	No	0.7561	0.9142	0.6599	0.7209	0.7949	0.8846	0.61
5	Deep NN	Hereditary_Cancer	Single	Yes	0.7551	0.8718	0.6669	0.6271	0.9487	0.7885	0.51
6	Neural Net (MLP)	Hereditary_Cancer	Single	Yes	0.7525	0.8785	0.6683	0.6129	0.9744	0.7692	0.55
7	Neural Net (MLP)	Hereditary_Cancer	Single	No	0.7500	0.8937	0.6560	0.6316	0.9231	0.7981	0.36
8	Deep NN	Hereditary_Cancer	Single	No	0.7400	0.8501	0.6465	0.6066	0.9487	0.7692	0.60
9	LightGBM	All	Multi	No	0.7024	0.8696	0.5655	0.7116	0.6934	0.8682	0.60
10	LightGBM	All	Multi	Yes	0.6970	0.8736	0.5367	0.6116	0.8102	0.7586	0.50
11	XGBoost	All	Multi	Yes	0.6900	0.8586	0.5299	0.6350	0.7555	0.7962	0.53
12	XGBoost	All	Multi	No	0.6859	0.8549	0.5363	0.6786	0.6934	0.8459	0.58
13	Deep NN	All	Multi	Yes	0.6853	0.8411	0.5173	0.5957	0.8066	0.7432	0.42
14	LightGBM	General	Single	No	0.6828	0.8558	0.5314	0.6468	0.7231	0.8238	0.53
15	Neural Net (MLP)	All	Multi	Yes	0.6796	0.8524	0.5075	0.5743	0.8321	0.7106	0.51
16	LightGBM	General	Single	Yes	0.6711	0.8530	0.5055	0.5952	0.7692	0.7666	0.51
17	XGBoost	General	Single	Yes	0.6681	0.8444	0.4991	0.5789	0.7897	0.7437	0.52
18	Deep NN	PAH	Single	Yes	0.6667	0.7435	0.5586	0.8000	0.5714	0.9318	0.48
19	XGBoost	General	Single	No	0.6611	0.8406	0.4869	0.5607	0.8051	0.7185	0.46
20	Deep NN	General	Single	Yes	0.6506	0.8210	0.4698	0.5347	0.8308	0.6773	0.47
21	Neural Net (MLP)	General	Single	Yes	0.6454	0.8163	0.4611	0.5277	0.8308	0.6682	0.51
22	Neural Net (MLP)	All	Multi	No	0.6208	0.7975	0.4185	0.4639	0.9380	0.4914	0.38
23	Deep NN	All	Multi	No	0.6177	0.7859	0.4000	0.4924	0.8285	0.5993	0.54
24	Deep NN	General	Single	No	0.6090	0.7740	0.4228	0.4429	0.9744	0.4531	0.43
25	Neural Net (MLP)	General	Single	No	0.6035	0.7329	0.4149	0.4361	0.9795	0.4348	0.39
26	Neural Net (MLP)	PAH	Single	Yes	0.5789	0.6591	0.4123	0.6471	0.5238	0.8636	0.51
27	XGBoost	PAH	Single	Yes	0.5652	0.6331	0.3329	0.5200	0.6190	0.7273	0.30
28	LightGBM	PAH	Single	Yes	0.5600	0.6667	0.3065	0.4828	0.6667	0.6591	0.24
29	Deep NN	PAH	Single	No	0.5588	0.6764	0.2805	0.4043	0.9048	0.3636	0.52
30	LightGBM	PAH	Single	No	0.5352	0.6385	0.2222	0.3800	0.9048	0.2955	0.11
31	XGBoost	PAH	Single	No	0.5333	0.6613	0.4850	0.8889	0.3810	0.9773	0.67
32	Neural Net (MLP)	PAH	Single	No	0.5143	0.6017	0.1657	0.3673	0.8571	0.2955	0.39

4. Model Comparison Heatmap

Heatmap showing all 9 evaluation metrics across all 32 configurations, sorted by F1 score. Green indicates higher values, red indicates lower values.

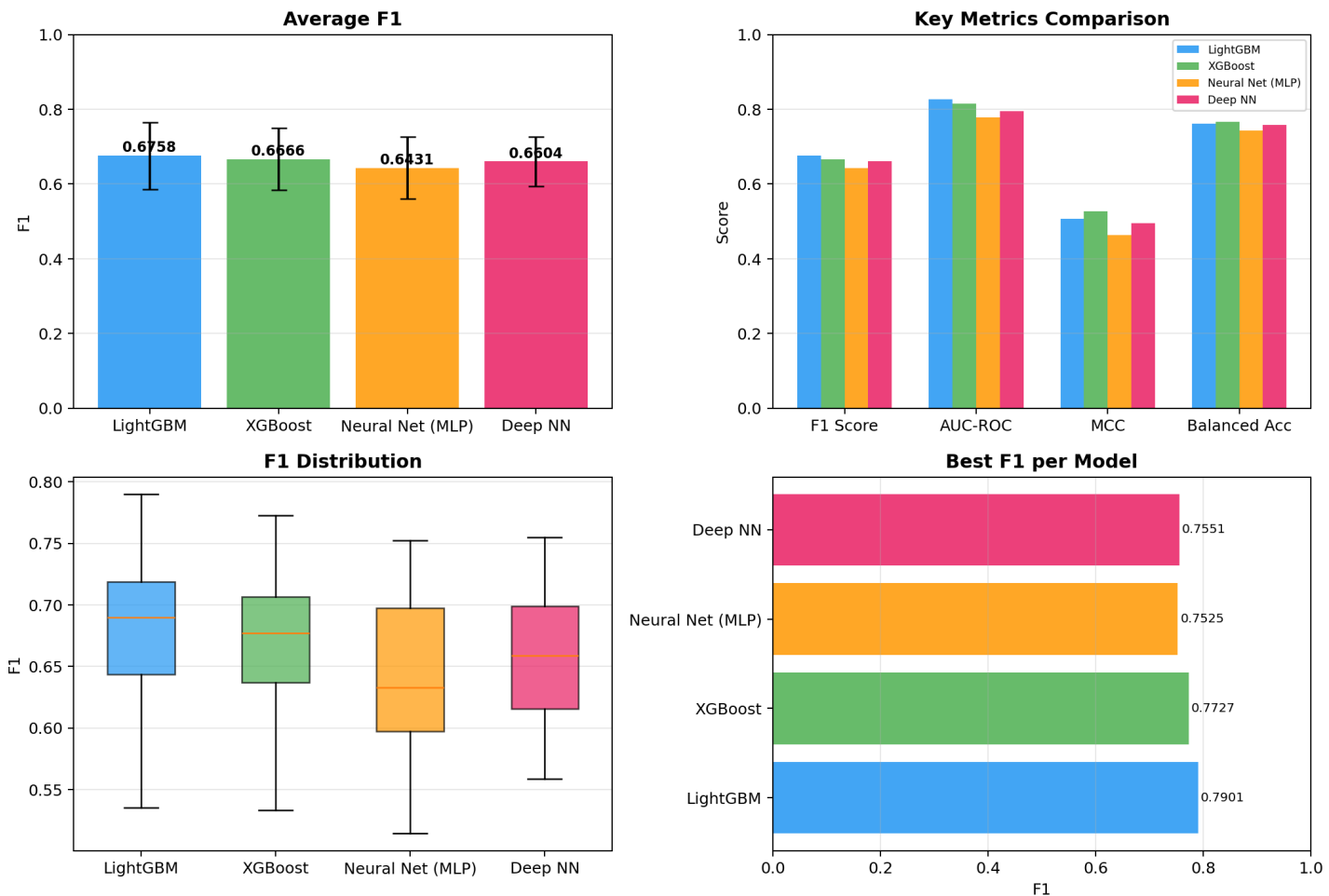


5. Model Type Analysis

LightGBM: Mean F1 = 0.6758 (+/- 0.0893) Range = [0.5352, 0.7901] (n=8)
XGBoost: Mean F1 = 0.6666 (+/- 0.0830) Range = [0.5333, 0.7727] (n=8)
Neural Net (MLP): Mean F1 = 0.6431 (+/- 0.0824) Range = [0.5143, 0.7525] (n=8)
Deep NN: Mean F1 = 0.6604 (+/- 0.0664) Range = [0.5588, 0.7551] (n=8)

Ranking: LightGBM (0.6758) > XGBoost (0.6666) > Deep NN (0.6604) > Neural Net (MLP) (0.6431)

Model Type Analysis



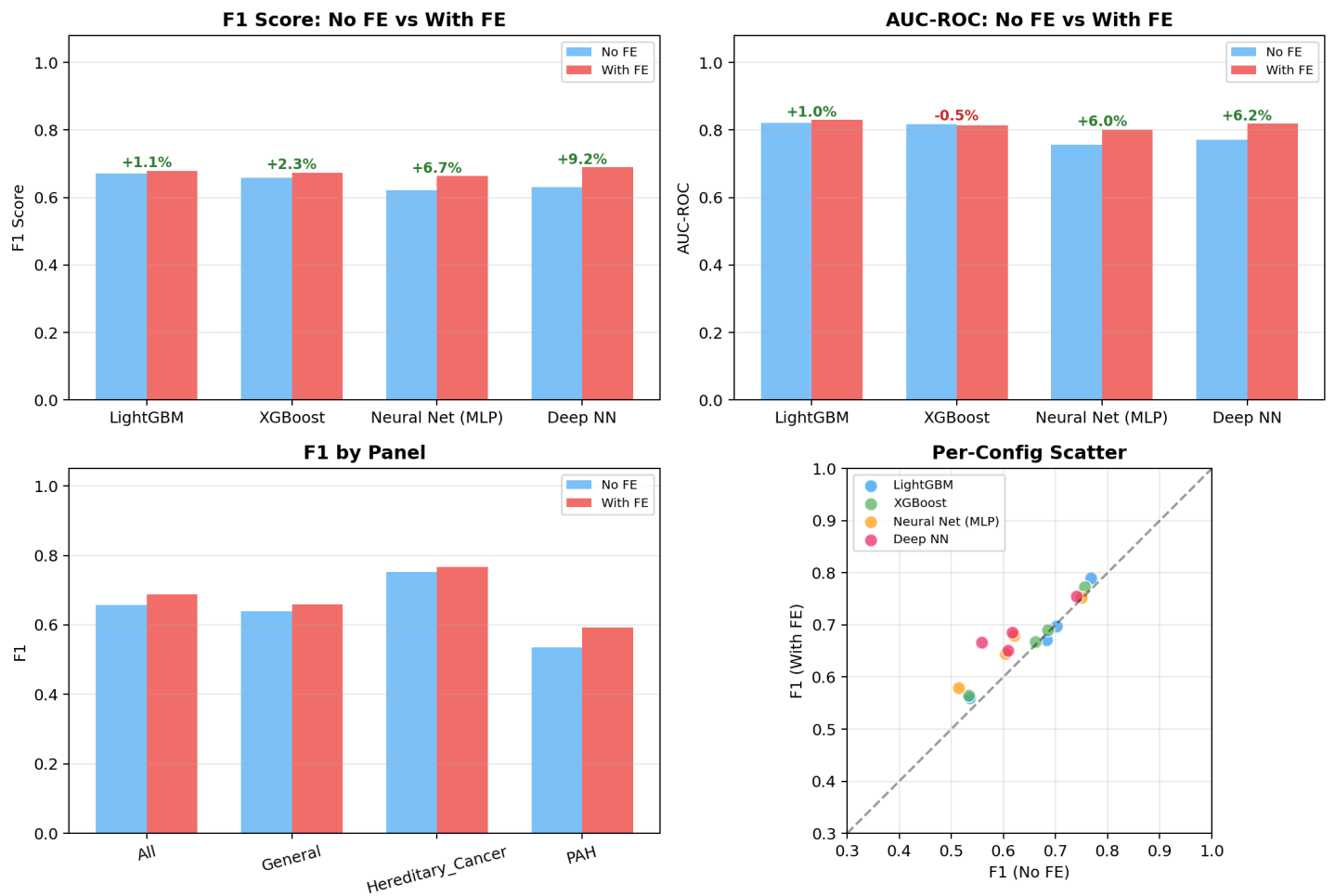
6. Feature Engineering Impact

No-FE avg F1: 0.6461 (~755 feat: 99 original + 656 k-mer)
 With-FE avg F1: 0.6768 (~57 feat: bioinformatics-derived)
 Difference: +0.0306 - Effect: IMPROVED

Per-model FE impact:

LightGBM: 0.6720 -> 0.6796 (+0.0076)
 XGBoost: 0.6591 -> 0.6740 (+0.0149)
 Neural Net (MLP): 0.6221 -> 0.6641 (+0.0420)
 Deep NN: 0.6314 -> 0.6894 (+0.0580)

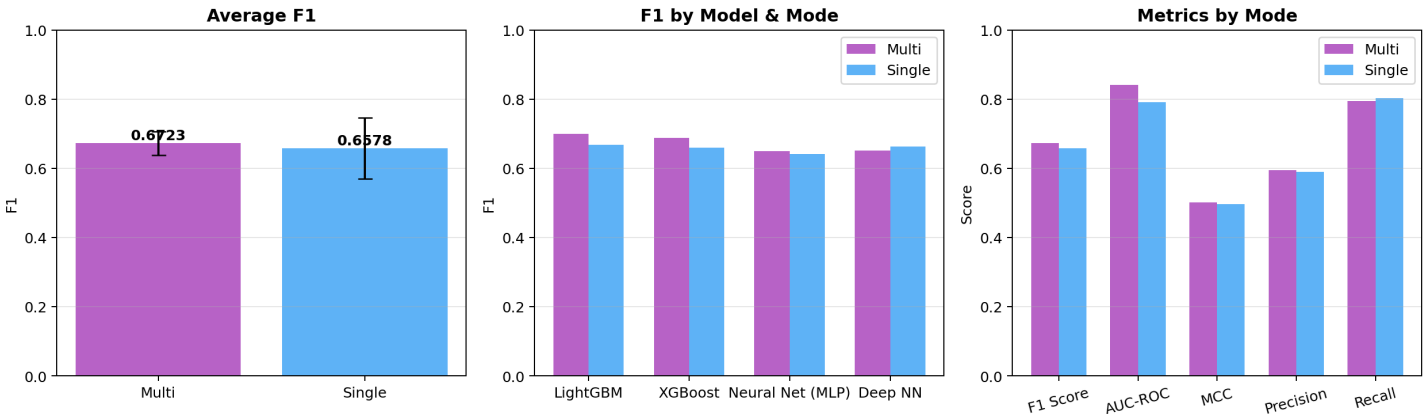
Feature Engineering Impact Analysis



7. Single-Panel vs Multi-Panel

Single-panel avg F1: 0.6578 (n=24)
Multi-panel avg F1: 0.6723 (n=8)
Difference: -0.0145

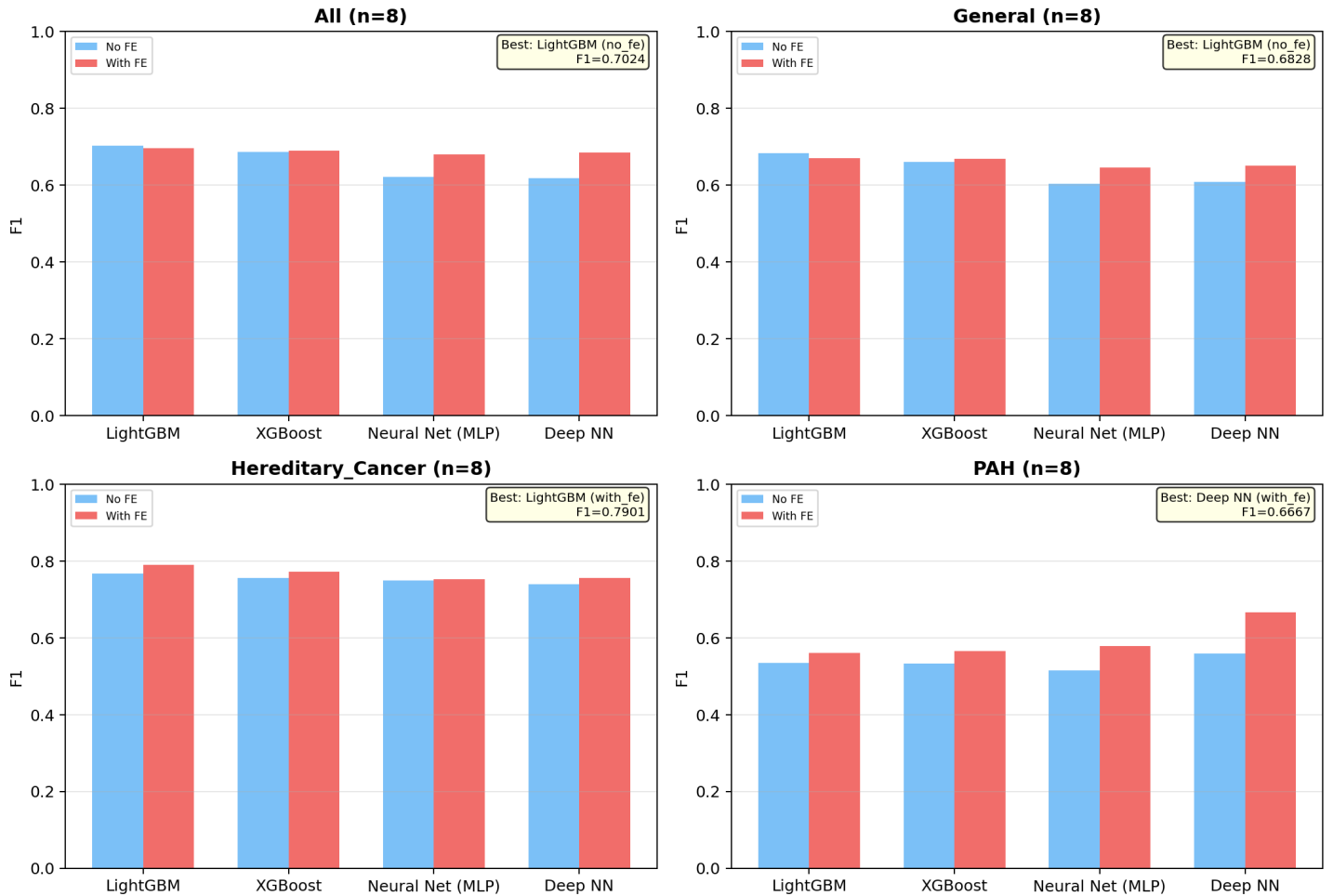
Single-Panel vs Multi-Panel



8. Panel-Specific Performance

All: Best = LightGBM (no_fe, multi_panel) F1=0.7024 AUC=0.8696 MCC=0.5655
 General: Best = LightGBM (no_fe, single_panel) F1=0.6828 AUC=0.8558 MCC=0.5314
 Hereditary_Cancer: Best = LightGBM (with_fe, single_panel) F1=0.7901 AUC=0.9283 MCC=0.7083
 PAH: Best = Deep NN (with_fe, single_panel) F1=0.6667 AUC=0.7435 MCC=0.5586

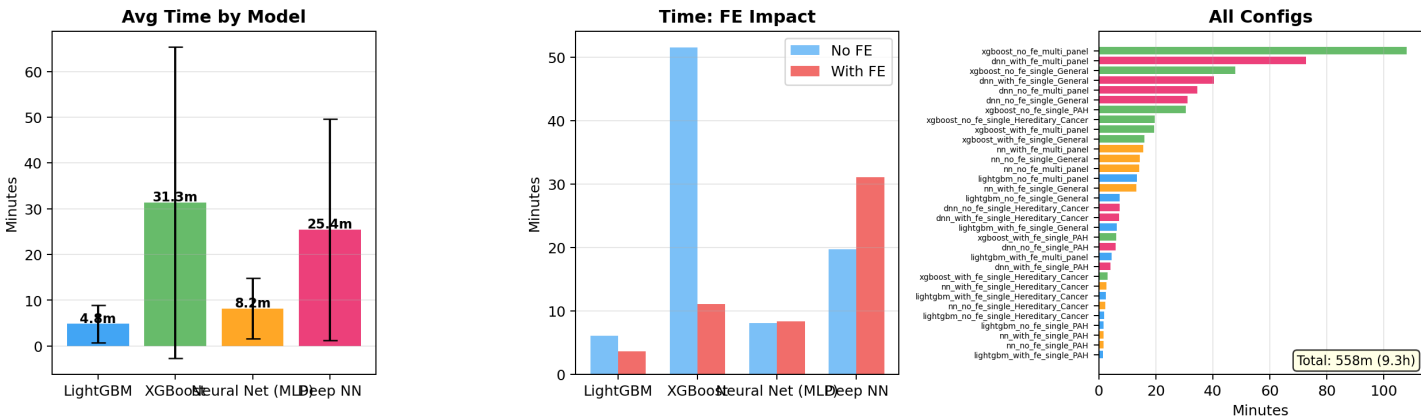
Panel-Specific Performance



9. Training Time Analysis

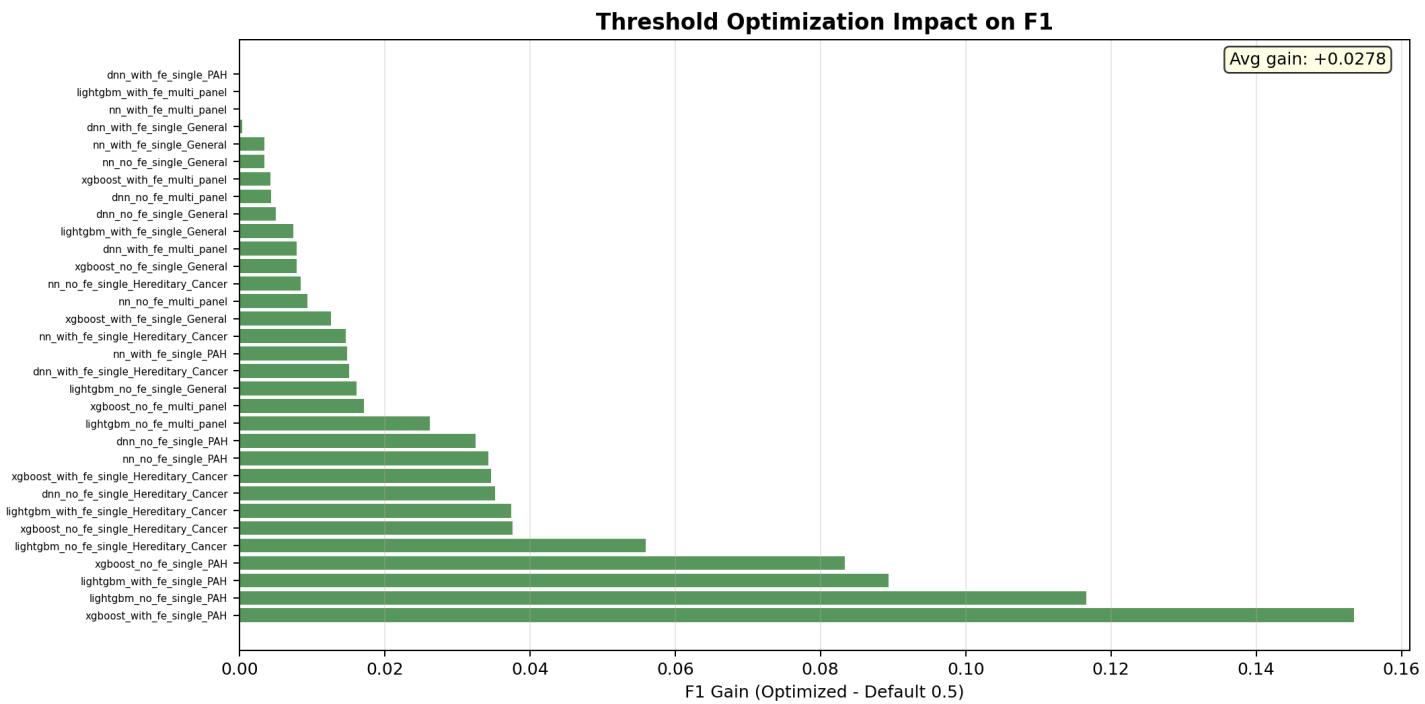
Total training time: 558 minutes (9.3 hours) for 32 configurations.

Training Time Analysis



10. Threshold Optimization Impact

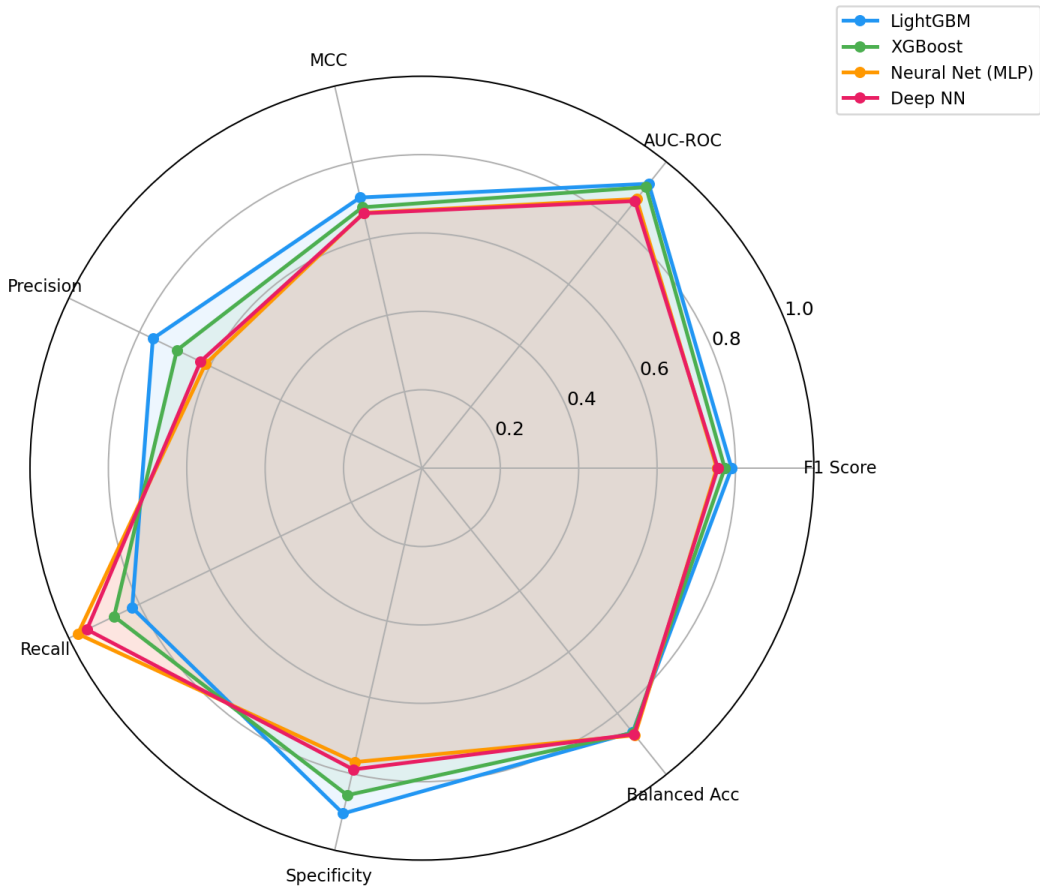
Post-hoc threshold optimization improved F1 by an average of +0.0278 compared to the default 0.5 threshold. 30 of 32 configurations benefited from threshold tuning.



11. Radar Chart - Best Config per Model

Visual comparison of the best configuration per model type across 7 key metrics.

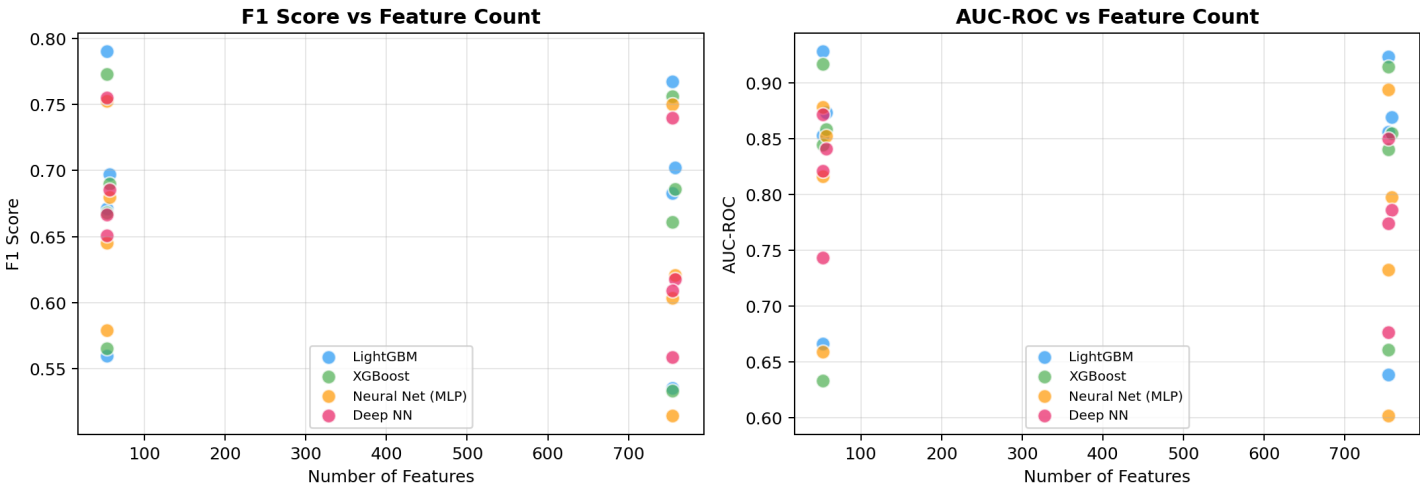
Best Config per Model - Radar Chart



12. Feature Count vs Performance

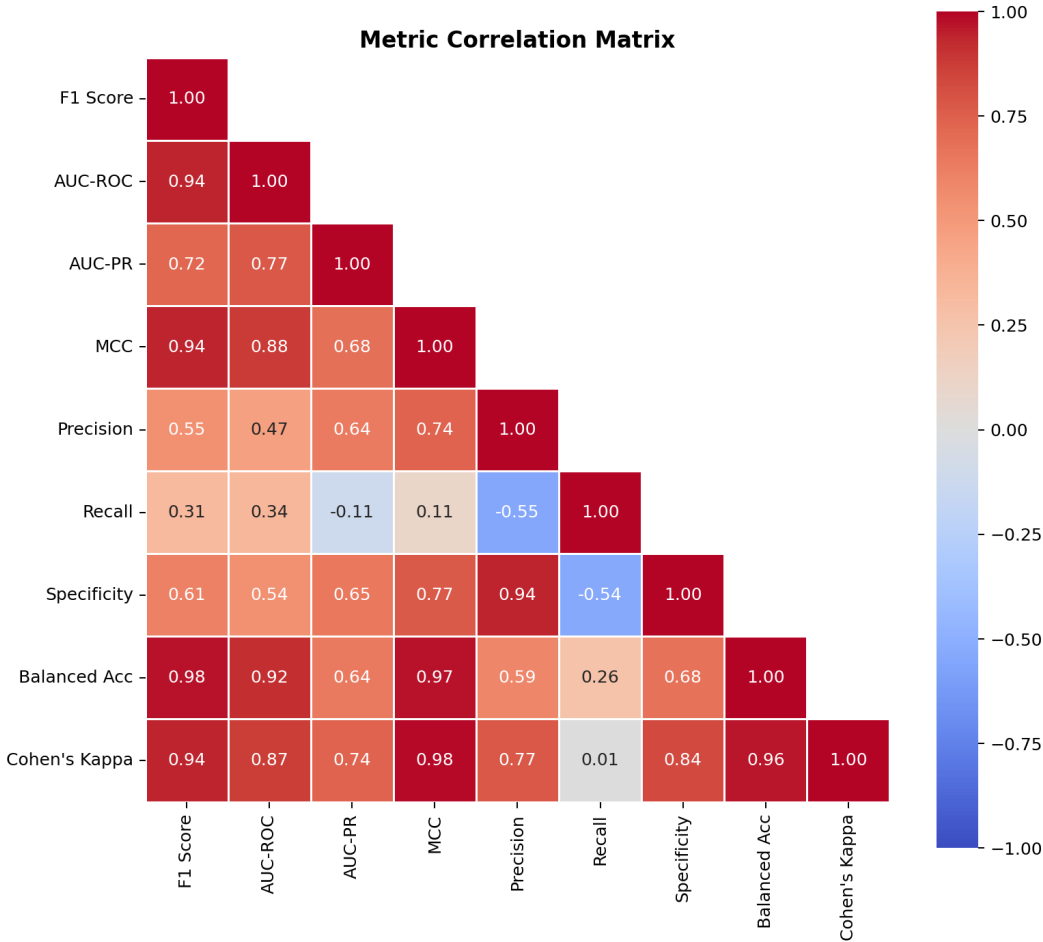
No-FE configurations use ~755 features (99 original + 656 k-mer from 6 sequence columns). With-FE configurations use only ~53-57 bioinformatics-derived features. This plot examines whether the large feature count leads to overfitting or genuine gains.

Feature Count vs Performance



13. Metric Correlation Matrix

Correlation between the 9 evaluation metrics across all 32 configurations. High correlation between F1 and MCC indicates consistent ranking. Lower correlation between Precision and Recall reflects the inherent trade-off.



14. Final Recommendation

BEST OVERALL CONFIGURATION

Config: lightgbm_with_fe_single_Hereditary_Cancer
Model: LightGBM
Panel: Hereditary_Cancer
Training Mode: Single-Panel
Feature Engineering: With Feature Engineering
Optimal Threshold: 0.620

Performance Metrics

F1 Score: 0.7901
AUC-ROC: 0.9283
AUC-PR: 0.7854
MCC: 0.7083
Precision: 0.7619
Recall: 0.8205
Specificity: 0.9038
Balanced Acc: 0.8622
Cohen's Kappa: 0.7074

Key Findings

1. Model Ranking (avg F1): LightGBM (0.6758) > XGBoost (0.6666) > Deep NN (0.6604) > Neural Net (MLP) (0.6431)
2. Feature Engineering: IMPROVED overall effect (+0.0306 avg F1). No-FE (0.6461) vs With-FE (0.6768). No-FE uses ~755 features (99 original + 656 k-mer from sequences), while With-FE distills information into ~57 bioinformatics features.
3. Training Mode: Single-panel (0.6578) vs Multi-panel (0.6723). Difference: -0.0145.
4. Threshold optimization provided an average F1 gain of +0.0278.

Clinical Deployment Considerations

- For screening (high sensitivity): lower threshold to ~0.3
- For confirmation (high specificity): raise threshold to ~0.7
- Current threshold (0.620) maximizes F1 (precision-recall balance)
- CFTR panel (100% pathogenic) excluded from single-panel evaluation

A. Appendix - Best Model per Panel

Panel	Model	FE	Mode	F1	AUC-ROC	AUC-PR	MCC	Prec	Recall	Spec	Thresh
All	LightGBM	No	Multi	0.7024	0.8696	0.7711	0.5655	0.7116	0.6934	0.8682	0.60
General	LightGBM	No	Single	0.6828	0.8558	0.6932	0.5314	0.6468	0.7231	0.8238	0.53
Hereditary_Cancer	LightGBM	Yes	Single	0.7901	0.9283	0.7854	0.7083	0.7619	0.8205	0.9038	0.62
PAH	Deep NN	Yes	Single	0.6667	0.7435	0.7181	0.5586	0.8000	0.5714	0.9318	0.48